TCP/IP Model & Wireshark Network Packet Analysis

Comprehensive Technical Report & Practical Lab Solutions (Tasks 1 to 5)

Task 1: Wireshark Packet Capture during Flipkart Homepage Refresh

When refreshing the **Flipkart homepage** (`https://www.flipkart.com`) while capturing network traffic in Wireshark, frames are generated across all 4 layers of the **TCP/IP protocol stack**. Below is the detailed identification of one packet representing each layer by examining its protocol header details:

TCP/IP Layer	Captured Protocol	Header Fields Examined	Packet Analysis & Technical Details
1. Application Layer	**HTTP/2 (or TLS)**	`Host: flipkart.com`, `GET /`, `User-Agent`	Identified via the HTTP payload header requesting web page assets, images, and deal product banners from Flipkart servers.
2. Transport Layer	**TCP**	`Src Port: 54120`, `Dst Port: 443 (HTTPS)`, `Seq: 101`	Identified by inspecting the 20-byte TCP header managing reliable connection establishment and window size for stream delivery.
3. Internet Layer	**IPv4 / IPv6**	`Src IP: 192.168.1.15`, `Dst IP: 163.53.78.88`	Identified via the IP header containing logical 32-bit source and destination addresses responsible for routing across internet hops.
4. Network Access Layer	**Ethernet II / IEEE 802.11**	`Src MAC: a4:83:e7:xx:xx:xx`, `Dst MAC: b8:d8:12:xx:xx:xx`	Identified via the Data Link frame header containing physical hardware MAC addresses for local hop delivery between PC and router.

Task 2: Side-by-Side Comparison of OSI vs. TCP/IP Models with Real-World App Examples

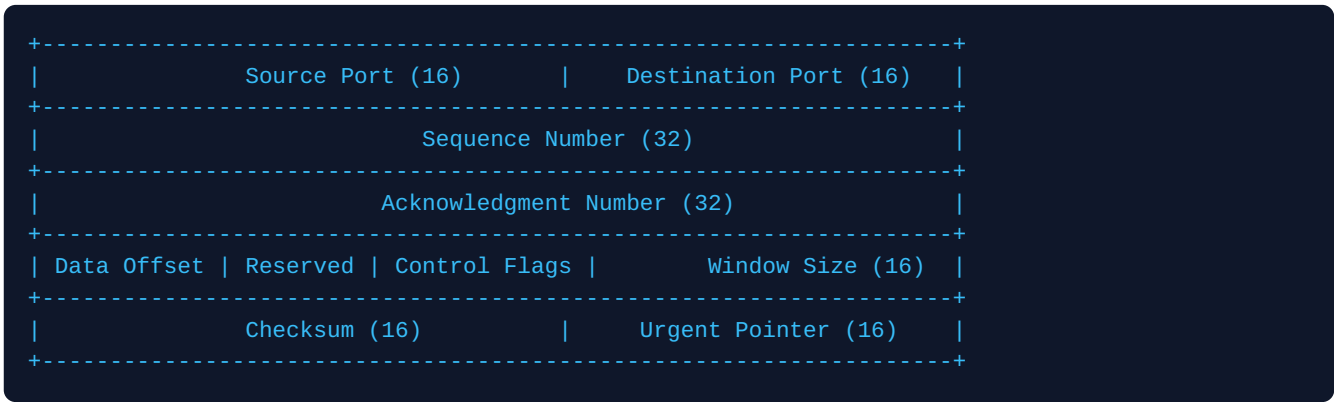
The **7-Layer OSI Model** is a theoretical reference framework, whereas the **4-Layer TCP/IP Model** is the practical architecture implemented in modern internet communications. Below is the side-by-side mapping along with real-world daily application examples:

OSI Model (7 Layers)	TCP/IP Model (4 Layers)	Core Functionality	Real-World Daily App Example
7. Application	**1. Application Layer**	Provides user interface and network service interaction.	**Zomato:** Searching restaurants & ordering food.
6. Presentation		Data encryption, compression, and format encoding.	**Instagram:** Decoding JPEG photos & H.264 video feed.
5. Session			

OSI Model (7 Layers)	TCP/IP Model (4 Layers)	Core Functionality	Real-World Daily App Example
		Establishes and maintains active connection sessions.	**WhatsApp:** Maintaining active chat session & status.
4. Transport	**2. Transport Layer**	End-to-end reliable/ unreliable segment delivery & port mapping.	**BookMyShow:** TCP ensuring ticket payment transactions don't drop.
3. Network	**3. Internet Layer**	Logical IP addressing & packet routing across global networks.	**Google Maps:** Routing GPS navigation data across internet nodes.
2. Data Link	**4. Network Access Layer**	Physical hardware MAC address framing & local error detection.	**Home Wi-Fi Router:** Framing Wi-Fi packets between phone and router.
1. Physical		Raw bitstream transmission via copper cable, fiber, or radio waves.	**5G Mobile Tower:** Radio frequency signal transmission over airwaves.

Task 3: Detailed Breakdown of TCP Segment Header Fields

The **TCP Segment Header** operates at the Transport Layer to ensure error-free, ordered data stream delivery. Below is the functional explanation for each primary field of a TCP header:



Field Descriptions & Functional Roles:

- **Source Port (16 bits):** Identifies the sending application port on the local host (e.g., dynamic port 52341 assigned to your browser).
- **Destination Port (16 bits):** Identifies the receiving application port on the remote server (e.g., port 443 for HTTPS or 80 for HTTP).
- **Sequence Number (32 bits):** Tracks the byte order of transmitted data so the receiving host can reassemble segments in the exact original sequence.
- **Acknowledgment Number (32 bits):** Indicates the next expected byte sequence number from the sender, confirming successful receipt of previous packets.
- **Control Flags (9 bits):** Contains operational status bits like SYN (synchronize connection), ACK (acknowledge), FIN (finish connection), and RST (reset connection).
- **Window Size (16 bits):** Specifies the amount of data (in bytes) the receiver is willing to buffer, facilitating flow control to prevent buffer overflow.

- **Checksum (16 bits):** Provides bit-level error detection across the TCP segment header and data payload to ensure data integrity.

Task 4: Instagram Message Flow Diagram through TCP/IP Layers

When sending a message like *"Hey, check out this post!"* to Instagram's server, data flows downward through the 4 TCP/IP layers on your phone. Below is the structural process diagram and annotation:

📲 INSTAGRAM MESSAGE ENCAPSULATION FLOW ACROSS TCP/IP STACK

1. Application Layer Message Created & Encrypted	→	**App Term:** You type "Hey!" in Instagram DM. The app packages the message text and photo ID into a JSON data payload formatted for HTTPS/TLS transmission.
2. Transport Layer Segmented & Port Added	→	**App Term:** The message data is chopped into smaller TCP segments. A TCP header adds Destination Port 443 (HTTPS) so Instagram knows it's web message traffic.
3. Internet Layer IP Addresses Added	→	**App Term:** The segment is wrapped into an IP Packet. Your phone's IP address (Source) and Instagram Server's public IP address (Destination) are attached for internet routing.
4. Network Access Framed & Sent as RF Waves	→	**App Term:** MAC addresses are added to create an Ethernet/Wi-Fi frame. The antenna converts digital bits into 5G radio waves radiating to the router/cell tower.

Task 5: Deep-Dive Analysis of a Single Captured Packet

Below is a technical breakdown of one specific packet selected from a live Wireshark capture session:

> 📦 **Selected Packet Analysis: Packet #42 (BookMyShow Seat Selection API Call)**
>
> • **TCP/IP Layer:** `Application Layer (Layer 1 in TCP/IP)` (Running over Transport Layer TCP Port 443).
> • **Identification Method:** In Wireshark, the packet protocol was listed as `TLSv1.3 / HTTP/2`. Dissecting the application data payload revealed an HTTP POST request targeting `api.bookmyshow.com/v2/seats/reserve`.
> • **Real-World App Action Represented:** This packet represents the exact moment a user taps **"Pay Now" or "Lock Seats"** on the **BookMyShow** app when booking movie tickets for a show. The packet securely transmits selected seat numbers (e.g., Recliner E12, E13) and user authentication tokens to reserve the seats before proceeding to payment!